



Viscous Damping Device for Parkinson's Tremors

ME574 Group 3

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Introduction & Problem

Tremors:

- Parkinson's disease causes involuntary tremors, which severely impact quality of life

Budget Constraints:

- Tasked to develop an effective solution that will balance affordability and high-quality performance, aiming to keep production cost low without sacrificing effectiveness

Mobility:

- Key requirement for our device is to not interfere with patients' mobility, but rather enhance by providing subtle and non-intrusive tremor relief



Solutions

- Pugh Matrix
 - Our group used a decision matrix to evaluate potential solutions based on criteria such as cost-effectiveness, ease of use, and manufacturability
 - Fluid damping was chosen as the most feasible option
- Patents
 - Patent research has been conducted to ensure our design is innovative and avoids infringement, while also being a marketable device



Active Solutions

- Our Solution:
 - Our product utilizes a fluid-based damping system designed to significantly reduce wrist tremors by offering a controlled, resistive force that stabilizes the user's wrist during movement.
- Fluid Damping Mechanism:
 - The fluid damping mechanism absorbs and controls the motion of the wrist, counteracting tremors in a smooth, efficient manner.



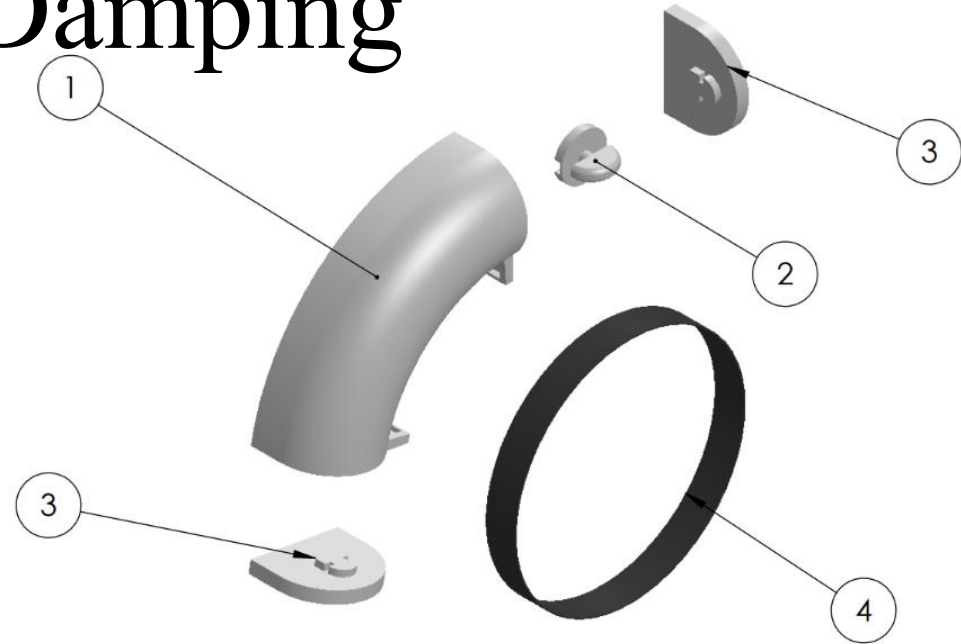


How to Use the Product

- User Instructions:
 - Step 1: Secure the device on the wrist and upper forearm using adjustable straps
 - Step 2: Wear throughout daily activities to reduce tremor impact
 - Step 3: Plug into charging port when battery is low

Fluid Damping

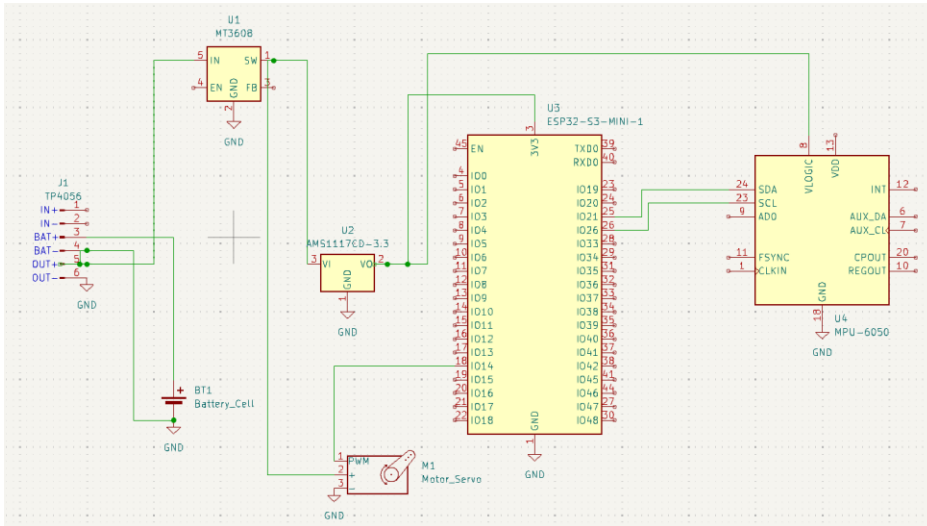
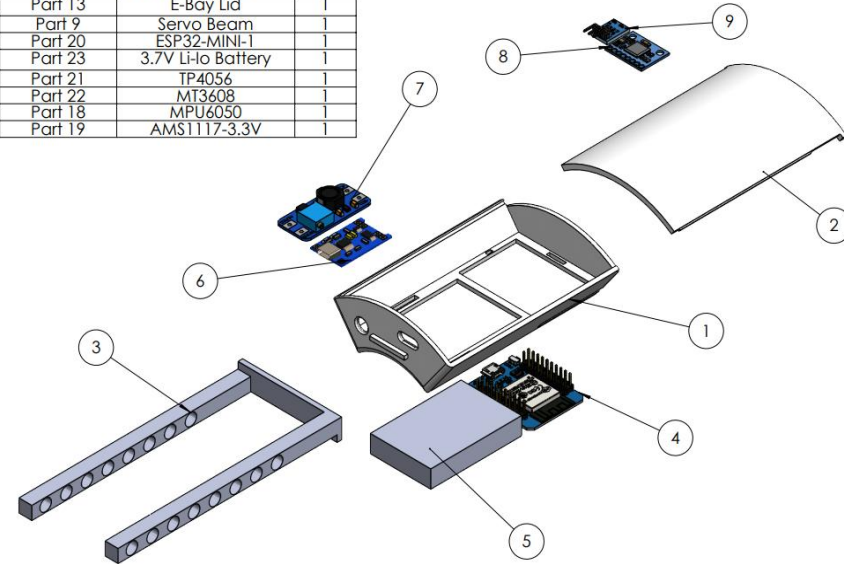
- Fluid Damping Principle:
 - The fluid inside the device works by offering resistive force that stabilizes the wrists motion. This mechanism provides an effective counterforce against the involuntary tremors.



ITEM NO.	PART NUMBER	DESCRIPTION
1	Part 5	Fluid resevoir, and beam rail
2	Part 6	Fluid Baffle with magnet
3	Part 7	Fluid resevoir cover
4	Part 16	Forearm Strap

Electrical System

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Part 12	E-Bay Case Assembly	1
2	Part 13	E-Bay Lid	1
3	Part 9	Servo Beam	1
4	Part 20	ESP32-MINI-1	1
5	Part 23	3.7V Li-Io Battery	1
6	Part 21	TP4056	1
7	Part 22	MT3608	1
8	Part 18	MPU6050	1
9	Part 19	AMS1117-3.3V	1





Electrical System

- Powering the Device:
 - The device is powered by a small, rechargeable battery. The low energy consumption ensures a long-lasting power supply with minimal charging frequency
- Sensor Integration:
 - Built-in IMU sensor detects tremor intensity and adjusts the fluid damping force according, providing an adaptive solution for varying levels of tremors



Manufacturability

- Injection Molding:
 - Parts will be manufactured using injection molding for precision and cost effectiveness, ensuring both quality and scalability.
- Purchased Parts:
 - Electronic components such as the sensors and batteries will be sourced from trusted suppliers to maintain consistency and performance.

Bill of Materials

Part Number	Description	Quantity	Cost Per Unit	Injection Tooling Cost
5	Fluid Housing and Rail	1	\$0.77	\$5605.00
6	Baffle	1	\$0.26	\$5348.00
7	Fluid Cap	2	\$0.37	\$4726.00
8	Magnet Holder	1	\$0.25	\$3897.00
9	Servo Beam	1	\$1.12	\$7460.00
10	Wrist Beam	1	\$0.42	\$6078.00
11	Telescoping Pin	1	\$0.13	\$4265.00
12	E-bay Case	1	\$0.59	\$7285.00
13	Ebay Lid	1	\$0.60	\$7409.00

Bill of Materials

Part Number	Description	Quantity	Cost Per Unit
14	Wrist Strap	1	\$0.99
15	Forearm Strap	1	\$2.99
16	Magnet	2	\$0.16
17	Servo	1	\$2.59
18	MPU6050	1	\$3.00
19	AMS1117-3.3	1	
20	ESP32-MINI-1	1	
21	TP4056	1	
22	MT3608	1	
23	3.7V Li-Io Battery	1	
24	Wires	15	
25	Canola Oil	1	\$2.39

Manufacturing Improvements

- Accessibility:
 - Final design feature telescoping beam to fit most adult arm sizes.
 - Final design has less weight to reduce interruption of ADLs.
- Optimized Component Size:
 - Product is cheaper to injection mold due decreased part footprint.
 - Final design utilizes around $\frac{1}{4}$ the material of initial design.





Cost and Market Potential

- Production Cost Per Unit:
 - Each unit is estimated to have a variable cost of \$17.16 covering materials, assembly and testing. Injection tooling is estimated to have a fixed cost of \$52,000.
- Sales Projection:
 - Based on the growing population and increasing prevalence of Parkinson's disease, we project 2000 units sold in the first 2-3 years.



Breakeven Point

- Revenue:
 - Estimated price of our product is \$70.
 - At 1000 units sold, the total revenue would equal \$70,000.
- Breakeven Point:
 - The breakeven point is expected to be reached after approximately 1000 units are sold, covering the development and initial production costs.



Social Impacts

- Social Benefits:
 - Improves independence for Parkinson's patients, reducing stigma associated with visible symptoms.
- Broader Impact:
 - Enhances quality of life for a growing segments of the populations.
 - Contribution to the medical assistive technology industry.

Future Testing

- Testing to be done to this product to ensure it can be made:
 - Longevity testing
 - Additional accessibility testing
 - Additional interest testing
- Future financial concerns
 - Product may not be financially viable after new tariffs.
 - Less interest in product with increase in prices.





Conclusion

- Key Points:
 - Device addresses tremor reduction in Parkinson's patients.
 - Meets standards, sustainable and offers strong market potential.

References

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